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阅读理解（说明文） How Does GPS Know Exactly Where You Are?

① Every morning, when your parents open the map app on their phone to check the traffic, when a delivery rider finds your small compound gate without asking, when a high-speed train running across the frozen north-eastern plain knows its speed within one kilometre per hour, when a group of lost hikers in the mountains of Yunnan sends out their precise coordinates to the rescue team — all of these small and big moments share the same quiet technology: **GPS**, which stands for **Global Positioning System**. Most middle school students use it every day without ever asking the simple question: **how does a small phone in your pocket know, within three metres, exactly where you are standing on the round Earth?** This short article explains, in simple words, the **four small steps** GPS uses.

② **Step 1 — The satellites above your head.** Roughly **20,200 kilometres above the Earth**, about 31 small satellites, each about the size of a family car, circle our planet twice a day. They do not belong to any single country's mobile phone company; they were built and launched by the United States Department of Defense, and since the year 2000 they have been freely available for **civilian use** anywhere on Earth. Each satellite carries something almost unbelievably precise: an **atomic clock**, accurate to **one second in three million years**. At all times, every satellite is whispering one short radio message to the Earth below: *"I am Satellite No. X, and my atomic clock now reads precisely 15:32:04.87320."* These whispers travel at the speed of light (about **300,000 kilometres per second**) and take only about **67 milliseconds** to reach your phone on the ground.

③ **Step 2 — Your phone secretly listens to four of them.** Inside every ordinary smartphone today is a small, low-power GPS receiver chip, about the size of a rice grain. This chip does **not** send any signal to the satellites — GPS is entirely a **listening** technology, not a speaking one. When you open a map app, your phone's GPS chip silently listens to as many satellites as it can clearly hear — usually between **five and twelve** at once — and chooses the **four with the clearest signal**. **Why four?** Because three satellites can locate your flat position on Earth (latitude and longitude), and the **fourth satellite** is needed to calculate your **altitude** (how high above sea level you are), and also to correct tiny clock errors inside your phone. Without the fourth, GPS would think you are on the floor even if you were on the 18th storey.

④ **Step 3 — The distance from time.** Here is the most elegant idea in the whole system. When your phone reads the message *"I am Satellite No. X, time 15:32:04.87320,"* it compares that satellite's time stamp with the current time inside the phone itself, and calculates the **tiny time delay** between them — perhaps 67.42 milliseconds. Since radio

signals travel at a fixed known speed (the speed of light), **time delay can be turned into distance**: 67.42 ms × 300,000 km/s = about **20,226 kilometres** of distance from that satellite. Your phone does this calculation **four times**, once for each satellite, giving it four precise distances to four known positions in the sky. If you imagine drawing four invisible spheres — one around each satellite, each with its own radius — the **single small point** where all four spheres overlap is exactly where your phone is standing on Earth. This geometric trick is called **trilateration** (多点定位), and the whole calculation is done inside your phone in **less than a second**.

⑤ **Step 4 — Small corrections.** A raw GPS reading would only be accurate to about **10 to 15 metres**, because the atmosphere slightly slows down the radio signals as they pass through its upper layers, and tall buildings can bounce signals around. To fix this, two small corrections are added. First, every modern phone compares the GPS reading with another quiet helper called **A-GPS** (Assisted GPS), which uses the nearby mobile-phone tower to correct small atmospheric errors. Second, in many countries including China, phones also quietly listen to a second satellite family called **BeiDou** (北斗), China's own satellite navigation system with more than 30 satellites. With both systems listening at the same time, most modern phones in a Chinese city now reach an accuracy of about **three metres**, which is why your map app can tell the difference between the north and south exits of a subway station.

⑥ Three small side facts worth remembering:

> **A.** GPS does not use any of your phone's mobile data or Wi-Fi. It works in the deep mountains, on a fishing boat, and inside a desert tent — anywhere under the open sky.
> **B.** GPS does **not** work well indoors under thick concrete ceilings; the weak radio signal cannot pass through many layers of steel and stone. That is why your map app often says *"signal searching"* inside a subway tunnel.
> **C.** GPS is **one-way**. The satellites never know that you are listening, and they never know your name, your age, or what you are doing. Whatever private information is shared on your phone is shared only by the **apps using the GPS reading**, not by the GPS system itself.

⑦ So the next time your parent's map app gently whispers, *"In 200 metres, turn left,"* you can remember, quietly, the small miracle behind that voice: four tiny satellite whispers from 20,000 kilometres up, each delayed by about 67 thousandths of a second, all arriving at the rice-grain chip in your phone, which in less than one second draws four invisible spheres in the sky and finds the single small point where they all meet — and that point is the one, exact, three-metre patch of Earth where you happen to be standing. **GPS is one of the most elegant examples of maths and physics quietly helping ordinary people find their way home.**

注释： GPS 全球定位系统；atomic clock 原子钟；civilian use 民用；altitude 海拔；trilateration 三/四边定位；BeiDou 北斗

根据短文内容，选择最佳答案。

1. What is Step 1 of how GPS works, and what do the satellites do?

A. The satellites are small and only fly 200 metres above the Earth every hour.
B. Roughly 20,200 kilometres above Earth, about 31 small satellites — each about the size of a family car — circle the planet twice a day; they were built and launched by the United States Department of Defense and have been freely available for civilian use anywhere on Earth since the year 2000; each satellite carries an atomic clock accurate to one second in three million years, and at all times every satellite whispers one short radio message — "I am Satellite No. X, and my atomic clock now reads precisely 15:32:04.87320" — which travels at about 300,000 km/s and takes only about 67 milliseconds to reach a phone on the ground.
C. The satellites send mobile phone text messages directly to everyone on Earth.
D. Each satellite carries a large wall clock and circles the Earth only once a year.

2. Why does a smartphone need to listen to exactly four satellites, and not three, according to Step 2?

A. Because the rice-grain chip inside a phone actively sends signals back to the satellites at full power.
B. Inside every ordinary smartphone today is a small low-power GPS receiver chip about the size of a rice grain that does not send any signal to the satellites — GPS is entirely a listening technology; when a map app opens the chip silently listens to as many satellites as it can clearly hear (usually between five and twelve) and chooses the four with the clearest signal; three satellites can locate flat position on Earth (latitude and longitude) and the fourth satellite is needed to calculate altitude (height above sea level) and to correct tiny clock errors inside the phone, without which GPS would think you are on the floor even if you were on the 18th storey.
C. Because three satellites can already give altitude perfectly and no fourth is ever necessary.
D. Because phones must talk to satellites with loud radio pulses each time a map is opened.

3. How does the phone turn satellite time signals into real distances in Step 3?

A. The phone guesses the distance randomly and hopes for the best.
B. When the phone reads the message "I am Satellite No. X, time 15:32:04.87320" it compares the satellite's time stamp with the phone's current time and calculates the tiny time delay between them — perhaps 67.42 milliseconds; since radio signals travel at the fixed known speed of light, time delay turns into distance (67.42 ms × 300,000 km/s ≈ 20,226 km from that satellite); the phone does this calculation four times, one for each satellite, giving four precise distances to four known positions in the sky; imagining four invisible spheres — one around each satellite — the single small point where all four overlap is exactly where the phone stands on Earth; this geometric trick is called

trilateration and the whole calculation is done inside the phone in less than a second.
C. The phone draws an invisible straight line to the centre of the Earth instead.
D. The phone sends the user's question directly to the satellites every second.

4. What two small corrections improve the raw GPS reading in Step 4?

A. No corrections are made and the phone simply shows a random location.
B. A raw GPS reading would only be accurate to about 10 to 15 metres because the atmosphere slightly slows the radio signals passing through its upper layers and tall buildings can bounce signals around, so two small corrections are added: first, every modern phone compares the GPS reading with another quiet helper called A-GPS (Assisted GPS) which uses the nearby mobile-phone tower to correct small atmospheric errors; second, in many countries including China phones also quietly listen to a second satellite family called BeiDou (China's own satellite navigation system with more than 30 satellites), and with both systems listening at once most modern phones in a Chinese city now reach an accuracy of about three metres, which is why a map app can tell the difference between the north and south exits of a subway station.
C. Corrections are done by asking nearby pedestrians for advice through voice commands.
D. BeiDou is only a Chinese weather satellite and has no connection with GPS correction.

5. According to paragraph ⑥, which of the following is TRUE about GPS?

A. GPS needs lots of mobile data or Wi-Fi and cannot work in remote areas.
B. GPS works perfectly under thick concrete ceilings and long subway tunnels.
C. GPS does not use any of your phone's mobile data or Wi-Fi and works anywhere under the open sky (deep mountains, a fishing boat, a desert tent); GPS does not work well indoors under thick concrete ceilings because the weak radio signal cannot pass through many layers of steel and stone, which is why a map app often says "signal searching" inside a subway tunnel; and GPS is one-way — the satellites never know that you are listening, they never know your name, age, or what you are doing, and whatever private information is shared on your phone is shared only by the apps using the GPS reading, not by the GPS system itself.
D. GPS satellites always know the user's name, age, and personal data.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段
【解析】"20200 km / 31 星 / 美军 2000 年起民用 / 原子钟 300 万年 1 秒 / 67 ms"直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段
【解析】"米粒芯片 / 仅监听不发 / 选最清 4 星 / 三星平面 / 四星海拔 + 钟校"直引。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段
【解析】"时延 × 光速 = 距离 / 4 个球交 1 点 / trilateration / 1 秒内"直引。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段
【解析】"原始 10–15m / A-GPS 基站 / 北斗 30+ 星 / 同听 3m 精度 / 地铁南北口"直引。A、C、D 均错。故选 B。

5.
【答案】C
【题型】细节理解题
【定位】第⑥段
【解析】"不需流量/WiFi / 开阔天空皆可 / 厚混凝土 / 单向不知用户"直引。A、B、D 均错。故选 C。